

Python Random Number Generator

Back-to-Basics Series

Python's Random Number

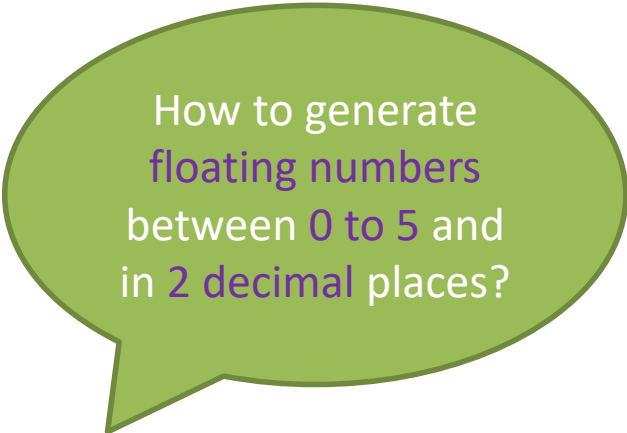
The random numbers generated by Python's `random` module is not truly random, it is **pseudo-random**; they are generated using a formula and hence they are deterministic.

If a **seed** value is provided, then the numbers generated will depend on the seed value; Otherwise, the system's current time will be used as the seed value to generate the random numbers.

1. Generate Random Number

Without using `seed(30)`, run the following code twice, and observe the two lists of random numbers generated.

```
from random import *  
  
seed(30)                # optional, can use other value.  
  
lst = []  
  
for i in range(10):  
    x = random()  
    lst.append(x)  
  
print(lst)
```

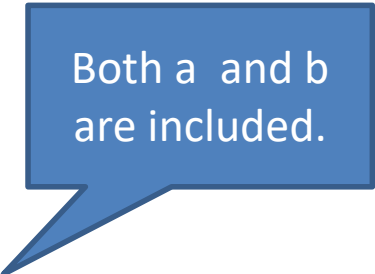


How to generate
floating numbers
between 0 to 5 and
in 2 decimal places?

2. Generate Random Integer

Without using `seed(30)`, run the following code twice, and observe the two lists of random numbers generated.

```
from random import *  
  
seed(30)                # optional, can use other value.  
  
lst = []  
  
for i in range(10):  
    x = randint(a,b)    # [a,b]  
    lst.append(x)  
  
print(lst)
```

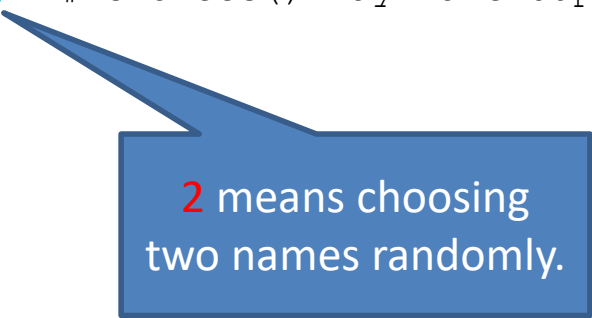


Both a and b
are included.

3. Random selection(s) From a List

Run the following code a few times to observe different selections of single or paired students generated.

```
from random import *  
  
lst = ['Adam', 'Bob', 'Charles', 'Daniel']  
  
single = choice(lst)  
  
pair = sample(lst, 2) # choices() may have duplicates  
  
print(single)  
  
print(pair)
```



2 means choosing
two names randomly.

4. Generate Normally Distributed Numbers

The following code will generate 1000 normally distributed random numbers:

```
from random import *  
  
lst = []  
  
for i in range(1000):  
    x = normalvariate(50,7)  
    lst.append(x)  
  
print(lst)
```

Mean = 50

Standard Deviation = 7

How to find the
`mean` and `standard
deviation` of the data
in a list? ?

The End
